

SIKKIM DATA DOSSIER

Everything pulled, derived, or generated for the landslide early-warning pipeline so far — every dataset, why it's there, what's in it, where it came from, and what's still open. Written to be read cold.

Prepared **2026-09-01** Scope **Feature 1 (susceptibility) + erosion side-model**

Pilot state **Sikkim** Scripts **01-10**

Overview / Role status / Dataset inventory / Findings & corrections /
What's left

01 SESSION OVERVIEW

The onboarding guide scoped this build to two features built genuinely real: a susceptibility model + GIS heatmap, and a rainfall alert + citizen reporting loop. This session covered the Data/GIS half of Feature 1 end to end — terrain, the landslide inventory, bias-controlled negative sampling, and a full training table — then went past the original scope to build the input layer set for a soil-erosion model, since bedrock lithology (blocked) turned out not to be the right input for that use case anyway.

Everything below is real: every download hit a live external source, every derived layer traces back to those downloads through a documented script, and every number quoted (record counts, class distributions, sanity-check stats) came from actually running the pipeline, not from copying the source docs. Two genuine errors in the team's own onboarding guide were caught and corrected along the way — see [Findings](#).

02 ROLE A STATUS

11

DATASETS COMPLETE

9

DERIVED / GENERATED
LAYERS

2

BLOCKED (EXTERNAL
FACTORS)

1531

ROWS HANDED TO ROLE B

WHAT'S ACTUALLY DONE

Every "core model input" the onboarding doc and dataset checklists call for is complete except lithology: boundary, DEM, slope, aspect, drainage density, land cover, the landslide inventory, bias-matched negative samples, and the joined training table are all sitting in data/processed/, sanity-checked against known Sikkim geography, and ready for Role B. Most dashboard context layers (roads, villages, hospitals/schools, population) are also done. Past the original scope, a full soil-property stack and a computed RUSLE erodibility factor exist for the erosion side-model.

WHAT'S BLOCKED, AND WHY THAT'S NOT A GAP IN THE WORK

Lithology and building footprints are blocked by things outside this pipeline's control — two GSI portals down, Overpass public infrastructure overloaded — not by unfinished effort. Each was tried multiple ways (see [dataset table](#) and [What's left](#)) before being parked, matching the team's own "time-box it, ship v1 without it" rule.

03 DATASET INVENTORY

Every dataset now in data/: what it is, why it's there, what's inside it, and exactly where it came from.

CORE TERRAIN

DATASET	TYPE	WHY IT'S USED	FIELDS / CONTENT	SOURCE
Sikkim boundary	• DOWNLOADED	Clip boundary for	shapeName, shapeISO,	geoboundaries DM1

DATASET	TYPE	WHY IT'S USED	FIELDS / CONTENT	SOURCE
		every other layer; defines the real analysis extent (27.08–28.13°N, 88.01–88.92°E) — the onboarding doc's stated extent was checked against this, not assumed.	shapeGroup, shapeType, geometry	
DEM tiles (×2)	• DOWNLOADED	Base elevation surface. Two tiles, not one — the doc's claim that a single tile covers Sikkim was wrong; the state's northern tip crosses into a second tile.	Single-band elevation (m), 30m, EPSG:4326	Copernicus Global Open Data
DEM, clipped + reprojected	• DERIVED	Analysis-ready elevation grid, aligned in EPSG:32645 (UTM 45N) to match every other layer in the pipeline.	Elevation (m), nodata = -9999, ~29m pixels	scripts/01_1
Slope	• DERIVED	Primary landslide predictor — steeper ground fails more. Landslide points measured at 33.0°	Degrees, 0–80° range	scripts/01_1 (Horn's method)

DATASET	TYPE	WHY IT'S USED	FIELDS / CONTENT	SOURCE
		median vs. 24.4° for matched negatives.		
Aspect	• DERIVED	Compass direction the slope faces — affects rainfall exposure and moisture retention.	Degrees, 0–360°	scripts/01_ (Horn's met

LANDSLIDE INVENTORY & SAMPLING

DATASET	TYPE	WHY IT'S USED	FIELDS / CONTENT
Landslide inventory (positive samples)	• DOWNLOADED	Labeled ground-truth landslide locations — the positive class. Pulled live (768 records), not from the 904-page PDF report the doc describes; the portal exposes a queryable ArcGIS FeatureServer instead.	Slide_No, State, District, Slide_Name, NH_SH_Location, Latitude, Longitude, Material_Involved, Movement_Type, History_Date, Activity (~150 raw GSI fields in the full record)
Roads	• DOWNLOADED	Dashboard context layer, and the anchor for two checks: independently verifying the inventory's road-proximity reporting bias (measured 97.0% within 100m, doc claims 96.1%), and generating negative	osm_id, highway (road class) 7357 segments, drivable classes only

DATASET	TYPE	WHY IT'S USED	FIELDS / CONTENT
		samples that share that bias.	
Negative samples	● GENERATED	The non-landslide class. Not downloadable from anywhere — algorithmically placed to match the positive samples' road-proximity distribution (median 6.0m vs. 6.7m, p90 24.6m vs. 29.3m) so a classifier can't just learn "near a road."	Longitude, Latitude, Distance_To_Road_M — 768 points

LAND COVER & DRAINAGE

DATASET	TYPE	WHY IT'S USED	FIELDS / CONTENT	SOURCE
Land use / land cover	● DOWNLOADED	Vegetated vs. bare/cut-slope feature. Distribution checks out against known Sikkim geography: 51% tree cover, 13% snow/ice, 10% moss/lichen alpine tundra.	Single-band class code, 10m 11 classes (10 Tree cover...100 Moss/lichen)	ESA WorldCover 20
Drainage density	● DERIVED	Proximity to streams — wet, undercut ground fails more. Landslide points sit measurably closer to streams (263m median) than negatives (344m), a real, plausible	Stream mask + distance-to-stream (m) range 0–1629m	scripts/06_derived (pysheds: fill dir → accumulat

DATASET	TYPE	WHY IT'S USED	FIELDS / CONTENT	SOURCE
		geomorphological signal.		

SOIL & EROSION (SIDE-MODEL)

DATASET	TYPE	WHY IT'S USED	FIELDS / CONTENT	SOURCE
Soil clay / sand / silt / organic carbon	DOWNLOADED	GSI lithology sources are down, and every global fallback tried (GLiM, Macrostrat) came back too coarse to vary at all within Sikkim. For erosion specifically, soil texture + organic carbon is the actually-correct input anyway (RUSLE's K-factor) — and this has real spatial variance (std dev 89 for clay), unlike the lithology fallbacks.	4 single-band layers, 0–5cm depth, 250m, value in g/kg (÷10 = %)	ISRIC SoilG
RUSLE K-factor (soil erodibility)	DERIVED	The actual erosion-model input: how easily surface soil detaches and washes away. Computed with the	Unitless, range 0.000–0.492, median 0.130 (lit. typical range	scripts/10

DATASET	TYPE	WHY IT'S USED	FIELDS / CONTENT	SOURCE
		standard Williams (1995) EPIC formula from the four soil layers above.	0.02–0.65)	

DASHBOARD CONTEXT

DATASET	TYPE	WHY IT'S USED	FIELDS / CONTENT	SOURCE
Villages	• DOWNLOADED	Location context for the dashboard.	osm_id, place, name 120 points	OSM via Overpass
Hospitals / clinics / schools	• DOWNLOADED	Emergency-response prioritization layer.	osm_id, amenity, name 106 points	OSM via Overpass
Population	• DOWNLOADED	Impact / prioritization overlay. Fetched at 1km, not 100m — the 100m server doesn't support range reads and 753MB wasn't worth it for an optional layer. Sanity-checked: 678,797 est. 2020 population vs. 610,577 in the 2011 census.	Single-band population count, 1km 2020 model estimate	WorldPop, Global 2
Buildings	• BLOCKED	Optional per every team doc. Failed the same way across 3	–	scripts/07c_fet

DATASET	TYPE	WHY IT'S USED	FIELDS / CONTENT	SOURCE
		different attempts (full query, rate-limit backoff, geographic quadrant split, including a single quarter of the state) — public Overpass infrastructure overload, not a query problem.		

MODEL HANDOFF

DATASET	TYPE	WHY IT'S USED	FIELDS / CONTENT	SOURCE
Training table	• DERIVED	The actual handoff to Role B (ML). One row per point, every layer above joined against each landslide/negative location.	lon, lat, elevation_m, slope_deg, aspect_deg, distance_to_stream_m, landcover_class, label 1531 rows (765 positive / 766 negative)	scripts/05_buil

BLOCKED

DATASET	TYPE	WHY IT MATTERS	WHAT WAS TRIED	SOURCES CHECKED
Lithology (bedrock geology)	• BLOCKED	Rock/soil type — a "core model input" per the doc, and a real predictor in the literature.	Both primary GSI portals unreachable (checked twice). PANGAEA's GLiM only freely offers a 0.5° grid (~55km cells — 2–4 values for the	bhukosh.gsi.gov.in , ngdr.mines.gov.in , GLIM/PANGAEA , Ma rostrat

DATASET	TYPE	WHY IT MATTERS	WHAT WAS TRIED	SOURCES CHECKED
			whole state). Macrostrat reachable but returns the identical classification at 5 points spanning the state – zero variance. OneGeology's WMS aggregator is itself unreachable.	

04 FINDINGS & CORRECTIONS

Things this pipeline caught that weren't in the source docs — the team's own culture calls for surfacing these, not hiding them.

The onboarding doc is wrong that one DEM tile covers Sikkim

The state's real bounding box (27.079–28.129°N) crosses into a second Copernicus tile above 28°N. Fixed by mosaicking both before clipping.

Un-set nodata silently corrupted the first DEM clip

The raw DEM tiles don't declare a nodata value. Clipping to Sikkim's polygon without one filled 33% of the raster (everything outside the state) with elevation 0 — indistinguishable from real low terrain, and it would have faked near-vertical cliffs in slope/aspect right at the boundary edge. Fixed with an explicit -9999 sentinel through the whole pipeline.

First negative-sampling pass had a systematic distance bias

Anchoring each candidate point to one road segment, then measuring distance to the *nearest* road afterward, pulled every measured distance down (negatives landed at roughly half the positives' distance at every percentile). Fixed by iterating against the true nearest-road lookup instead of a fixed anchor.

Road-proximity bias independently re-verified, not just copied from the doc

Measured 97.0% of landslide points within 100m of a road using freshly-pulled OSM data, against the doc's stated 96.1% — consistent. All 768 points also spot-checked (exhaustively, not just 15) as falling inside the Sikkim boundary polygon.

A second, related bias surfaced in land cover — flagged for Role B

Landslide points hit the "Built-up" land-cover class at 10.2% vs. only 5.1% for the bias-matched negatives. Negative sampling was only built to match road-proximity, not land-cover class, so this is likely the same reporting artifact leaking through a different feature. Worth checking feature importance before trusting `landcover_class` as a genuine signal.

05 WHAT'S LEFT, IN ORDER

The remaining work to close out Role A, ordered by what actually blocks the rest of the team.

1 Confirm the training table has actually reached Role B

Data prep is done — `training_table.csv` has everything the doc calls essential except lithology. The doc's own hard rule is a model training on real data by end of Day 2; nothing has been fit yet as of this handoff. Getting this file in front of whoever's doing ML is the single highest-leverage next action.

BLOCKS — THE WHOLE SUSCEPTIBILITY MODEL

2 Build the RUSLE LS-factor and C-factor

K-factor (soil erodibility) is done. LS (slope length-steepness) derives from the slope raster already on disk; C (cover management) derives from the land-cover classes already on disk, mapped to standard literature C-values per class. Both are same-day work — no new downloads.

UNBLOCKS — THE EROSION RISK MAP

3 Combine $K \times LS \times C \times R$ once Role C's rainfall erosivity (R-factor) lands

This produces the actual soil-loss estimate — a real, physically-based RUSLE output, not an unsupported ML claim (there's no labeled erosion ground-truth to train a classifier against, unlike the landslide inventory).

4

Sanity-check Role B's trained risk map against real terrain

Per the Data/GIS role's own Day 4 responsibility: does the "high risk" output actually sit on steep slopes near drainage lines? Flag it before it goes into the demo unexamined.

DEPENDS ON — ROLE B FINISHING A FIRST MODEL

5

Export terrain layers for the dashboard (Role D)

Slope, elevation, land cover, and the erosion factors need to reach the frontend in a renderable form (web tiles or simplified GeoJSON), not just as analysis GeoTIFFs.

BLOCKS — THE GIS HEATMAP DASHBOARD

6

Retry buildings when Overpass isn't overloaded

Optional and non-blocking — the script (`07c_fetch_buildings_chunked.py`) is ready, it just needs the public Overpass servers to not be under load. Worth one more attempt later, not worth waiting on now.

NON-BLOCKING — DASHBOARD EXTRA

7

Keep an ear out for lithology access

If anyone gets GSI portal access from a different network, or a teammate already has a geology file, it's still worth adding. Otherwise, this stays absent and gets stated plainly on the pitch's honesty slide — not implied as covered.

NON-BLOCKING — DOCUMENT EITHER WAY

8

Brief Role F (PM/Pitch) on everything in Findings

The 2-tile DEM correction, the re-verified road bias, the built-up-class flag for Role B, and the lithology gap all belong on the honesty slide before a jury asks. This pipeline already found and fixed a real methodological issue — that's a stronger signal than never having noticed it.

BLOCKS — Q&A PREP ACCURACY

Full technical detail, exact commands, and every dead end tried lives in data/PROVENANCE.md. Scripts referenced above are in scripts/01 through scripts/10. This dossier reflects the pipeline's state as of 2026-09-01 – re-run the relevant script and re-check this page if any upstream source changes.